

HQDFM Design for Manufacture(DFM) Report

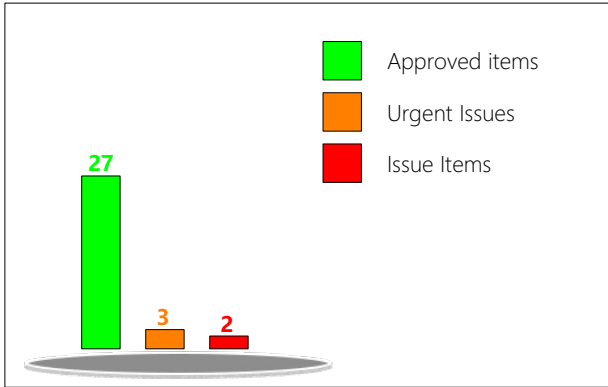
File name: 2026-06-23

Time: 2026-06-23 Layer count:4

PCB Thickness: 1.60

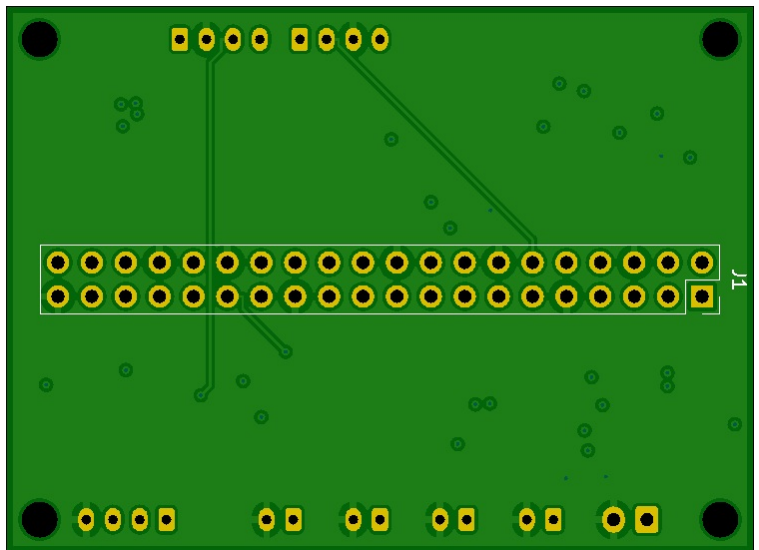
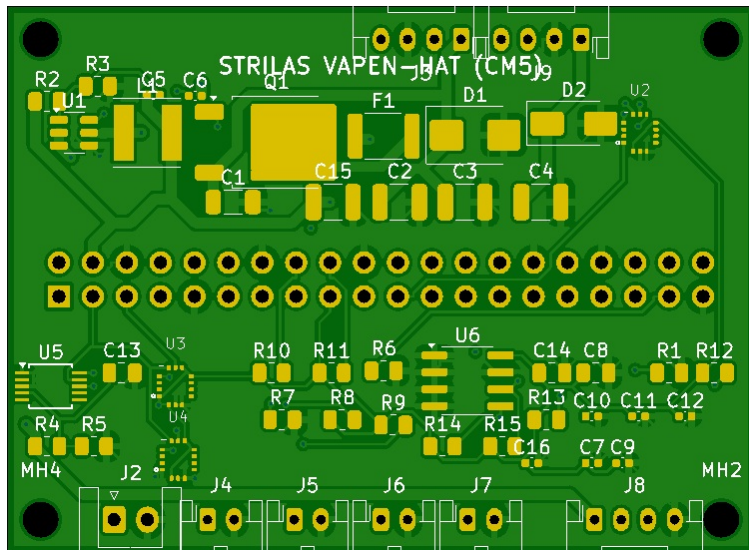
Quantity: 5

mm



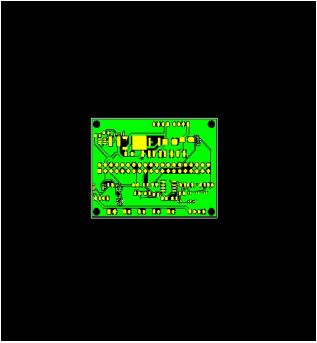
Basic Board Specs	Trace Width/Spacing	8.00/8.00mil
	Milling Density	88.0557m/m²
	Surface Finish Area	28.71%
	Test Point Count	201

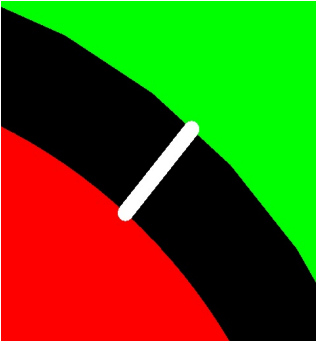
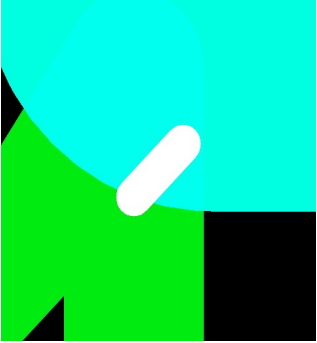
The file size is small, which can affect the surface mount assembly process. It is recommended to have a size larger than 7*7cm. You can optimize the size by adding a process edge or increasing the panelization.



Type	Category	No. of Checks	Result
PCB Trace Analysis	Open/Shorts (IPC)	1	Fail
	Signal Integrity	4	Fail 4
	Smallest Trace Width	1	Pass 4
	Smallest Trace Spacing	3	Pass 539
	SMD Pad Spacing	1	Pass
	Pad Size	3	Pass 5 , Fail 2
	Hatched Copper Pour	2	Pass
	Annular Ring Size	2	Pass 2
	Drill to Copper	5	Pass 246 , Fail 8
	Copper-to-Board Edge	2	Pass 11
	Holes on SMD Pads	4	Pass
PCB Drilling Analysis	Drill Diameter	8	Pass 9
	Drill Spacing	4	Pass 4
	Drill to Board Edge	4	Pass
	Drill Hole Density	1	Pass
	Special Drill Holes	2	Pass
	Drill Hole Errors	3	Pass
PCB Solder Mask Analysis	Solder Mask Dam	2	Pass 9
	Missing SMask Opening	1	Pass
	Solder Paste Area	1	Pass
PCB Silk Analysis	Silkscreen Spacing	1	Fail 8

PCBA Component Analysis	Component Spacing	1	Fail
	Comp.-to-Board-Edge	3	Fail
	Component Silkscreen Spacing	0	Fail
	Pad Count Mismatch	4	Fail
	Designator Length	0	Fail
	Double-sided Components	1	Pass
	Mid Mount Comp	1	Pass
	Comp Height Analysis	2	Pass
PCBA Pin Analysis	SMT Pad Analysis	8	Fail
	Through-hole Pins	10	Fail
PCBA Pad Analysis	Chip Pads	60	Fail
PCBA Fiducial Analysis	Fiducial Count	1	Fail
	Fiducial Analysis	4	Pass

ID	Check	Limits	Value	Issue	Image	Position	Qty	Level
1	Signal Integrity_Acute Angle Traces	-,,-	Error(s) detected	There is an acute angle connection mode in your design,which will affect the signal integrity of the product.it is recommended to adjust the acute angle position to arc or obtuse angle connection mode		161.45,-102.45	2	Warning
2	Pad Size_Short Pads	8,12,20	0.28 mm	Regular pads of size 0.28mm were detected in your design. This could result in overetching, detached pads and increased testing costs, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. The pads should be enlarged to at least 0.3 mm if possible.		152.00,-121.25	4	Warning

3	Drill to Copper_NPTH-to-Copper	8,10,12	0.25 mm	NPTH to copper spacing of 9.86mil was detected in your design. This could increase the risk of defects such as short circuits, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. It is recommended to increase the spacing to at least 12 mil.		125.46,-136.88	8	Warning
4	Silkscreen Spacing_Solder Mask-to-Silkscreen	4,5,6	Error(s) detected	For most factories, the minimum silkscreen to solder mask spacing requirement is at least 8 mil. Failure to meet the factory's requirements could result in part of the silkscreen being removed or being printed directly on the pads, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. Silkscreen to solder mask spacing of 0 mil were detected in your design. It is recommended to increase the spacing to at least 12 mil.		127.55,-106.16	8	Risk